

help the retailer perform an intelligent product allocation and inventory optimization.

The retailer can also come up with a dashboard on customer experience, by taking into account users spending time at a promotion zone, in the aisle and time taken to make a purchase.

For analysis of unstructured data, text mining algorithms can be used for automatic extraction of keywords. These include key features of the product, along with the models and brands retrieved from social media and blogs.

Results and discussion

Popular keywords in unstructured data: The word cloud in Figure 3 depicts the popular features and characteristics of bags extracted from social media and blogs..

Figure 4 shows the Kibana dashboard of the various types of bags, the popular characteristics from social media and in-store inventory levels.

A few analyses based on the comments received on social media and on blogs related to the bags are given below:

- People are more concerned about the product than the store they buy them in.
- Adjustable and detachable features like shoulder straps appear to be important for buyers.
- While colour, material/fabric and style are important, people are also conscious of pricing.
- Crown and crocodile print based designs are most commented on and popular.

Data insights

From the comments on the blogs, we get to know that print designs are one of the key elements that customers are looking at. However, bags of this design are not available in-store. Hence, it would be appropriate to stock these popular bag designs as soon as possible. These inputs can help retailers in the decision-making process related to product assortment and promotional offers. Similarly, new insights can be obtained by combining different data sources.

Amalgamating multi-structured data in an enterprise using open source technologies and methods can help an enterprise to gain insights into data and perform intelligent product allocation. It can also help sales personnel to send out effective messages to



Figure 3: Popular keywords from unstructured data



Figure 4: Kibana dashboard of bags

the customer. This can also help in improving product literature, highlighting the key features that customers are looking for.

The example from the retail industry are applicable to other industry verticals such as telecom, utilities and manufacturing. Specific use cases in telecom can be in customer churn risk prediction, improving the experience in customer care channels, real-time promotions, etc. Input data sources will be from social media, call logs, CRM and billing information. A solution that addresses these use cases can positively impact awareness of customer tastes and reduce the customer churn rate. 

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